L J UNIVERSITY
L J SCHOOL OF COMPUTER APPLICATIONS
SEMESTER 7
SUBJECT: MACHINE LEARNING
QUESTION BANK

Human Learning

Define the concept of human learning.

Describe the process of learning from expert guidance in human learning.

How does learning from knowledge guided by the expert differ from the other types of learning?

Give an example of how a human can learn a new skill through self-learning.

Machine Learning

Define machine learning.

How does machine learning differ from traditional programming?

What are the main types of machine learning?

Types of Machine Learning

What is supervised machine learning, and what are its key features?

Describe unsupervised machine learning and provide an example of its application.

Compare and contrast supervised and unsupervised machine learning.

Supervised Machine Learning

What is the difference between classification and regression tasks in supervised learning?

Give an example of a classification problem and a regression problem.

Describe a scenario where regression would be preferred over classification.

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Unsupervised Machine Learning

What is clustering, and how is it used in unsupervised learning?

How does association rule learning differ from clustering?

Provide an example of a real-world application of clustering.

Explain the concept of association rules with an example.

Applications of Machine Learning

Write and explain any five applications of Machine Learning in real life.

How is machine learning used in financial forecasting?

Provide an example of machine learning in the field of natural language processing (NLP).

Tools Used for Machine Learning

What role does Python play in machine learning, and why is it preferred?

Describe the purpose of Pandas in data analysis and machine learning.

How is Matplotlib used in the machine learning workflow?

Explain the function of sklearn (sci-kit learn) in machine learning.

What are the advantages of using Jupyter Notebook for machine learning projects?

Types of Data in Machine Learning

What are categorical and ordinal data, and how do they differ?

Define nominal data and give an example.

Explain the difference between structured and unstructured data.

What is the difference between primary and secondary data sources?

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Data Preprocessing

What is data preprocessing, and why is it essential in machine learning?

Describe the common steps involved in data preprocessing.

Data Remediation Techniques

What are some techniques for handling missing values in a dataset?

How can duplicate values be identified and removed from a dataset?

Define outliers with suitable example

Data Transformation

Explain label encoding and its purpose in machine learning.

Describe standard scaling and its significance in preprocessing.

Describe minmax scaling and its significance in preprocessing.

Feature and Model Selection

What is feature selection, and why is it important?

How does model selection differ from feature selection?

Write full form of PCA.

Independent Features vs Dependent Features

Define independent features and dependent features give proper example.

Training Dataset vs Testing Dataset

What is the purpose of a training dataset in machine learning?

Why is it important to have a separate testing dataset?

Explain the process of splitting data into training and testing dataset.

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Predicting Output and Accuracy Score

What is meant by predicting output in a machine learning model?

Write method to print accuracy score of the model using sci-kit learn.

K-Nearest Neighbors (KNN) Algorithm

Explain the K-Nearest Neighbors (KNN) algorithm with an example.

Sums of K-Nearest Neighbors.

Euclidean Distance

What is Euclidean distance, and how is it used in machine learning?

Provide a formula for calculating Euclidean distance.

Support Vector Machine (SVM) and Its Terminology

What is a Support Vector Machine (SVM), and what is its primary purpose?

Explain the concept of a hyper plane in SVM.

What are support vectors, and why are they important in SVM?

Describe the role of kernels in SVM.

List types of kernels in SVM.

Define Margin.

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Decision Tree

What is a decision tree, and how does it works?

Explain the concept of entropy in decision trees. How does it influence the decision-making process at each node?

What is information gain, and how is it used to construct a decision tree?

Find the entropy of whole dataset and entropy, information gain of outlook from the play tennis dataset.

Naive Bayes

Write the formula of Bayesian Theorem and explain Posterior, Prior and Likelihood.

Sums of Naïve Bayes.

Regression

Explain the different types of Linear Regression with proper diagrams.

Explain Simple Linear Regression with example.

Explain Multivariate Linear Regression with example.

Sums of Linear Regression

Model Performance Evaluation

Sums and formula to find accuracy score, precision score, recall score, f1 score from confusion matrix.

Clustering

Sums of KMean Clustering

Sums of KMedoids Clustering

Write Formula of Manhattan Distance.

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Association

Sums of Apriori Algorithm.

Formulas of Support, Confidence and Lift.